Advance Python Assignment

Topic-9: SQLite3 and PyMySQL (Database Connectors)

Q1. Explain SQLite3 and PyMySQL in Python.

Ans.
SQLite3 and PyMySQL are used in Python to connect with databases and perform operations like storing and retrieving data. SQLite3 is a simple, lightweight database that is built into Python and does not require any external setup. It is mainly used for small applications. PyMySQL is used to connect Python with MySQL databases and is suitable for large-scale applications. These tools help in managing structured data using SQL queries and are widely used in web and data-related projects.

Example:
```
import sqlite3

connection = sqlite3.connect("data.db")
print("Connected to database")
connection.close()
```

Q2. Explain how to execute SQL queries using SQLite3 in Python.

Ans.
In Python, SQL queries can be executed using SQLite3 by creating a connection and a cursor object. The cursor is used to execute SQL commands like creating tables, inserting data, and retrieving records. The execute() method is used to run SQL queries. After performing operations, commit() is used to save changes in the database. This allows Python programs to work with database data efficiently.

Example:
```
import sqlite3

conn = sqlite3.connect("data.db")
cur = conn.cursor()

cur.execute("CREATE TABLE IF NOT EXISTS user(id INTEGER, name TEXT)")
cur.execute("INSERT INTO user VALUES(1, 'Shivani')")
```

```
conn.commit()
conn.close()
```